UNIVERSITI TEKNIKAL MALAYSIA MELAKA

INDIVIDUAL LAB REPORT

PROGRAM

BITA GROUP 2 SEMESTER 1

ASSESSMENT

LAB 4

SUBJECT

BITP1133

LECTURER

DR ISRA GHANI

DATE OF SUBMISSION

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BIL	NAME	NO MATRIC
1	MUHAMMAD AMIRUL AZIZ BIN MUHAMMAD FIRDAUS	B032610858

Exercise 4

1) Write a C++ program that uses a loop to iterate through N = 1 to 8 cores for a program with 80% parallelizable code (f = 0.80) and 20% sequential code (0.20).

```
Cores (N)   Speedup
------------------------
    1       |   1.00x
    2       |   1.67x
    3       |   2.14x
    4       |   2.50x
    5       |   2.78x
    6       |   3.00x
    7       |   3.18x
    8       |   3.33x
```

2) Submit your C++ source code (.cpp) and a screenshot of the console output as your lab deliverable.

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    double f = 0.80; // 80% parallelizable code

    cout << "Cores (N)\tSpeedup" << endl;
    cout << "------------------------" << endl;

    for (int n = 1; n <= 8; ++n) {
        double speedup = 1.0 / ((1.0 - f) + (f / n));
        cout << "    " << n << "       |   "
             << fixed << setprecision(2) << speedup << "x" << endl;
    }

    return 0;
}
```